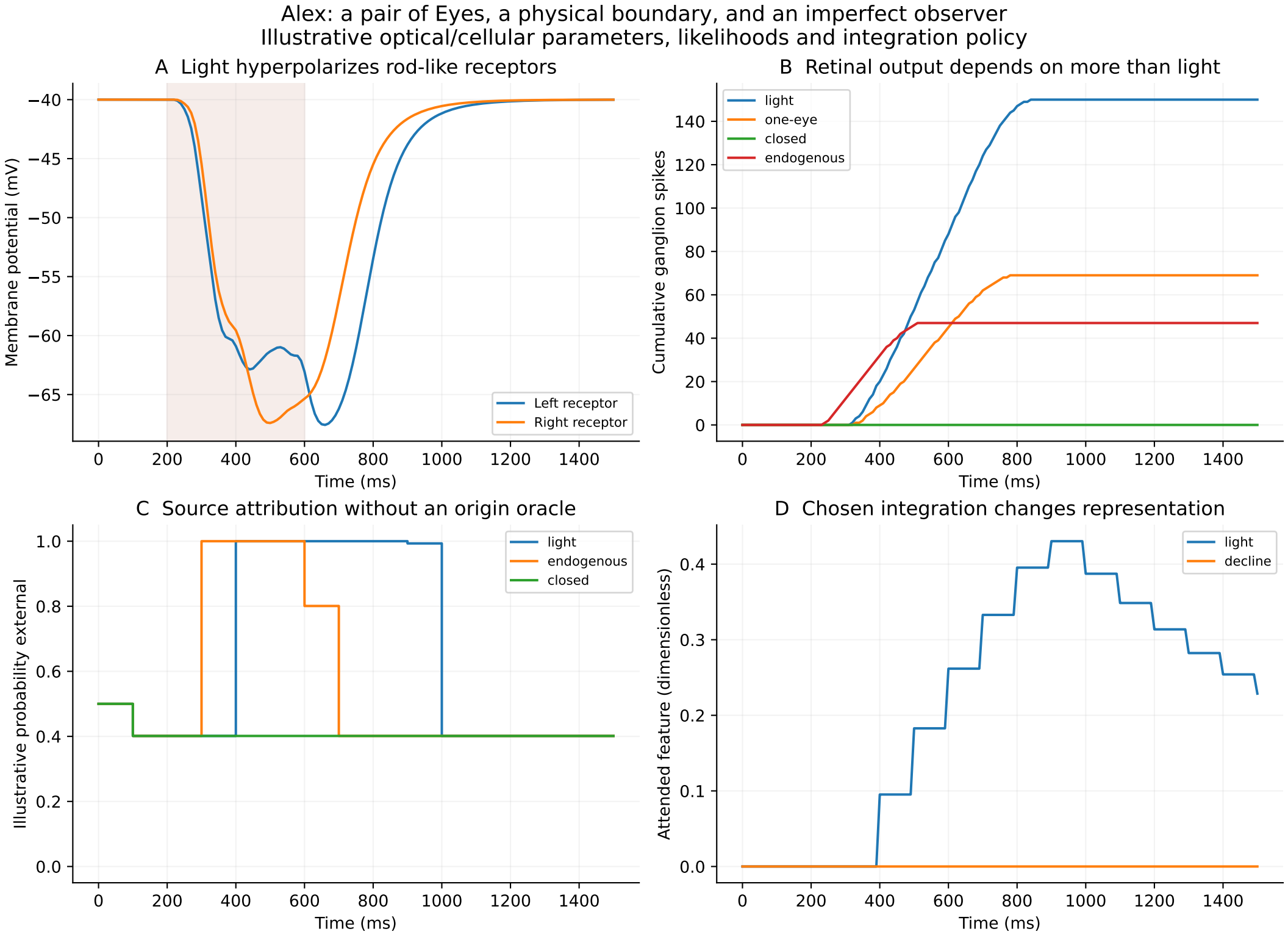
Alex: a pair of Eyes, a physical boundary, and an imperfect observer
Illustrative optical/cellular parameters, likelihoods and integration policy
A Light hyperpolarizes rod-like receptors
Membrane potential (mV)
−40
−45
−50
−55
−60
−65
Left receptor
Right receptor
0
200
400
600
800
1000
1200
1400
Time (ms)
B Retinal output depends on more than light
Cumulative ganglion spikes
light
one-eye
closed
endogenous
140
120
100
80
60
40
20
0
Time (ms)
C Source attribution without an origin oracle
Illustrative probability external
1.0
0.8
0.6
0.4
0.2
0.0
light
endogenous
closed
Time (ms)
D Chosen integration changes representation
Attended feature (dimensionless)
light
decline
0.4
0.3
0.2
0.1
0.0
Time (ms)